

ISHAAN NANDA

☎ +91 7496072116 | ✉ nandaishaan19@gmail.com | 🔗 LinkedIn | 🐙 GitHub | </> Codeforces

EDUCATION

Vellore Institute of Technology

Bhopal, India

Integrated Masters of Technology in Computer Science Engineering Aug. 2022 – May 2027 (Expected)
(Specialization in Computational and Data Science)

CGPA: 8.93/10.0

Relevant Coursework: Data Structures and Algorithms, Operating Systems, Machine Learning, Object-Oriented Programming, Probability, Statistics and Reliability, Linear Algebra

EXPERIENCE

Research Intern, Vikram Sarabhai Space Centre (VSSC), ISRO

Nov 2024 – Dec 2024

- Developed a **real-time object detection and tracking system** using **YOLOv8** and **DeepSORT** for space applications
- Integrated **DepthAI pipelines** for optimized frame acquisition from **OAK-D cameras**, achieving a processing time of **77ms per frame**
- Configured **OAK-D cameras** through a network switch, ensuring seamless multi-camera synchronization
- Enhanced object tracking accuracy by **fine-tuning DeepSORT parameters** for real-time applications
- Implemented custom preprocessing techniques to improve detection performance under varying lighting conditions

PROJECTS

Apple Plant Disease Detection System



Python, TensorFlow, Convolutional Neural Networks (CNNs), Image Classification

- Constructed a machine learning model to detect **10 common diseases** in apple plants with **82% accuracy**
- Utilized **CNNs for image classification**, training on a dataset of **18,000+ leaf images**
- Incorporated **data augmentation**, increasing the training dataset by **2x** and improving model robustness
- Developed a user-friendly interface, reducing disease identification time from **15 minutes to 2 minutes** per sample

Spatial Data Structure Performance Analysis for Restaurant Search



Python, Spatial Data Structures, Algorithm Optimization

- Implemented and analyzed **Linear Search, Grid-based Spatial Index, and R-tree** methods, achieving **85% faster query times** with R-tree implementation
- Created comprehensive test scenarios using **100+ randomized queries** across varying dataset sizes to evaluate efficiency
- Analyzed and documented specific performance metrics including time complexity, memory usage, and scalability for large-scale applications
- Developed technical documentation comparing implementation approaches and their practical applications in food delivery systems

TECHNICAL SKILLS

- Languages: Python, C++, Java, SQL
- Frameworks & Libraries: OpenCV, TensorFlow (basic knowledge)
- Concepts: Data Structures and Algorithms, Machine Learning (Basic)
- Developer Tools: Git

COMPETITIVE PROGRAMMING

Codeforces



- Became **Pupil** with a max rating of **1203**, demonstrating consistent problem-solving skills and algorithmic thinking
- Successfully solved **200+ algorithmic problems** spanning topics like dynamic programming, graph theory, and data structures
- Participated in **15+ rated contests**, showing steady rating improvement and ability to perform under time constraints

INTERESTS

Travelling, Fitness, Food, Puzzles